

sanity

A simple Typst package to find unreferenced figures, uncited sources and lost labels.

version 0.1.0, github.com/techgustavo/sanity



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Getting started

sanity analyzes your document after it has been compiled by Typst and reports unreferenced figures, uncited bibliography entries, as well as captions and labels that have gone missing. You add it as a show rule.

```
#import "@preview/sanity:0.1.0": *  
  
#show: sanity
```

With these lines, whenever there is something to report, a page is attached listing what was found, just like this:

```
sanity v0.1.0 3 warnings, 1 info
```

warning	equation <eq:capacity> is never referenced unreferenced-equation	page 1
warning	figure <fig:noise> is never referenced unreferenced-figure	page 1
info	figure <fig:noise> has no caption missing-caption	page 1
warning	bibliography entry "example2026" is never cited uncited-entry	

sanity added this page; report: none, or --input sanity=off, leaves it out.

What to read next

[The checks](#) lists every check (what it reports). [Configuration](#) and [Leaving something out](#) are how you change its behavior. [From the command line](#) and [In CI](#) read the findings without touching the document.

The package needs Typst 0.14 or newer, but the command line script needs 0.15 (where `typst eval` arrived).

The checks

Twelve checks run as soon as you apply the `show` rule. The other six are off until you ask for them with `checks: { "id": true }`.

An element is *only* reported as unreferenced if you gave it a label, so a figure with no label *cannot* be pointed at anyway.

Elements nothing points at

unreferenced-figure	default:	on	warning
----------------------------	----------	-----------------	----------------------

A labelled figure that no `@ref` and no `#link(<label>)` in the document points at.

unreferenced-table	default:	on	warning
---------------------------	----------	-----------------	----------------------

The same, for a figure whose `kind` is `table`.

unreferenced-listing	default:	on	warning
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The same again, for a figure whose `kind` is `raw`. A figure of some other kind is named after its `kind` in the message but reports under `unreferenced-figure`.

unreferenced-equation	default:	on	warning
------------------------------	----------	-----------------	----------------------

A labelled block equation that nothing points at.

unreferenced-footnote	default:	on	warning
------------------------------	----------	-----------------	----------------------

A labelled footnote that nothing points at.

unreferenced-heading	default:	off	warning
-----------------------------	----------	------------------	----------------------

A labelled heading that nothing points at. `off` by default.

Note: Elements Typst cannot number are never called unreferenced.

Labels and structure

duplicate-label	default:	on	error
------------------------	----------	-----------------	--------------------

One label attached to more than one element. Typst rejects any reference to a repeated label.

heading-level-skip	default:	on	warning
---------------------------	----------	-----------------	----------------------

A heading two or more levels below the one before it, like a `===` straight after a `=`.

reference-order	default:	off	warning
------------------------	----------	------------------	----------------------

A figure or table first mentioned out of numbering order (figure 2 discussed before figure 1). This is not a rule of documents at large, so it is off by default.

Captions

missing-caption

default: on warning

A *labelled* figure or table with no caption at all.

empty-caption

default: on warning

A caption with nothing in it.

missing-label

default: off warning

A captioned figure or table with no label. It is off by default because some documents caption figures they never intend to reference.

duplicate-caption

default: off warning

Two figures, or two tables, carrying the same caption.

The bibliography

uncited-entry

default: on warning

An entry in the bibliography data that nothing in the document cites. It can only run when sanity has that data (see [The bibliography](#)).

bibliography-not-checked

default: on info

Says that uncited-entry could not run, and names the file to pass so that it can.

Exemptions

orphaned-ignore

default: on warning

A `sanity-ignore` that matches nothing.

Bibliography keys count as matches, since `#sanity-ignore("styleguide")` is how you keep an entry you never cite.

A document that has to be tagged

missing-alt-text

default: off warning

An image with no `alt` text.

table-without-header

default: off warning

A table with no `table.header` row.

Configuration

```
#show: sanity.with(  
  checks: ("unreferenced-heading": true, "empty-caption": false),  
  severities: ("duplicate-label": "warning"),  
  bibliography: read("refs.bib"),  
  report: none,  
  strict: true,  
)
```

Argument	Default	What it does
checks	(:)	A dictionary keyed by check id.
severities	(:)	The same, with "info", "warning" or "error" as values.
bibliography	none	What read() returns for a .bib or .yaml file (or an array of them).
report	auto	auto appends the page and none leaves the document completely untouched.
strict	false	true fails the compilation on any warning/error

Nothing at info fails a build.

```
#show: sanity.with(severities: ("empty-caption": "info"), strict: true)
```

Leaving something out

Some figures are meant to be unreferenced, so you can exempt them:

```
#sanity-ignore(<fig:cover>, reason: "decorative")
```

It takes labels or plain strings, and any number of them, so it also works for labels built at compile time.

```
#for id in appendix-ids {  
  sanity-ignore(label("fig:" + id))  
}
```

To silence one check rather than everything about an element:

```
#sanity-ignore(<fig:map>, checks: "unreferenced-figure")
```

reason is just a note to whoever reads the source, if the exemption goes stale, the reason appears in the orphaned-ignore message, so you know which one to delete.

The bibliography

A Typst package cannot read your `.bib` file. But from the command line, `bin/sanity` reads the files your document names and hands them over.

Inside the document, pass the data yourself, like this:

```
#show: sanity.with(bibliography: read("refs.bib"))
```

`BibTeX` and `Hayagriva` files are both understood and an array covers a document with more than one.

```
#show: sanity.with(bibliography: (read("primary.bib"), read("extra.yml")))
```

From the command line

The `bin/sanity` script reports on a document without modifying it. It exits non-zero when it finds a warning or an error.

```
$ bin/sanity paper.typ
warning: figure <fig:latency> is never referenced [unreferenced-figure]
└─ page 4

sanity: 1 warning
```

You can download the script from GitHub and make it executable.

```
curl -sLO https://raw.githubusercontent.com/techgustavo/sanity/main/bin/sanity
chmod +x sanity
```

<code>--json</code>	Print the findings as a JSON array instead of a report.
<code>--version</code>	Print the package the script imports.
<code>--help</code>	Print the flags.

`--json` gives an array of dictionaries with the keys `id`, `severity`, `message`, `target`, `page` and `page-label`.

```
$ bin/sanity paper.typ --json
[{"id":"unreferenced-figure","severity":"warning", ..}]
```

In CI

Exits 1 on a warning or an error, 0 otherwise, and 2 when the document does not compile.

```
name: manuscript
on: [push, pull_request]
```

```
jobs:
  sanity:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: typst-community/setup-typst@v5
      - run: |
          curl -sSL0 https://raw.githubusercontent.com/techgustavo/sanity/main/bin/
sanity
    chmod +x sanity
    ./sanity paper.typ
```

strict: true setting causes the compilation itself fails, so typst compile gates the build and no script is needed. To get a PDF out of such a document anyway, pass `--input sanity=report`, or `--input sanity=off` for the untouched one.

Thanks for considering this package!
